

Report 19 Jan 2026

N=191

20% cases pet done before first surgery

30% cases are primary spondylodiscitis

- Primary 60
- Secondary: surgery 53
- Secondary: nonsurgery 52
- Recidive (Discitis) 24

New spinal focus after pet is rare: 3.6%

New non-spinal focus after pet is quite common: 18.3%

CRP correlates strongly with STAU infection (ca. +0.5)

HWS might be strongly associated with new non-spinal focus in PET (i.e. strong recommendation to do PET in HWS discitis)

Cervical Spine	PET: New Nonspinal Focus?	%
0	0	87.90%
0	1	12.10%
1	0	54.29%
1	1	45.71%

- Almost 3x more likely to find a nonspinal focus in cervical spine discitis with PET than in non-cervical spine.
- See analysis per focus type below.
- Investigate the relationships with: Microbiology

Patho is more reliable in detecting spondylodiscitis

Pathology	Negative	Positive
Microbiology		
Negative	27.27%	72.73%
Positive	5.83%	94.17%
All	11.59%	88.41%

- Microbiology pos, Pathology neg: 7 cases, mostly cutibacterium acnes (probably contamination) + Anderson Lesion + mibi not done
- Positive pathology with negative microbiology is very often (72%)

Tracer uptake is seen in 78.5% of cases

- Predicting Pathology from PET: Sensitivity 95.8%, Specificity 26.3%
- Predicting Microbiology from PET: Sensitivity 95.8%, Specificity 13.6%

- In literature Specificity 95% / sensitivity 96%: check how they calculated (Smids et al, others?)
- Subgroup analysis
 - Region
 - PET spinal TE always positive in HWS discitis (31/191 cases)
 - Seen always in MRT? Usefulness of PET?
 - Region: BWS, Sensitivity: 0.98, Specificity: 0.40
 - BWS more specific
 - Region: LWS, Sensitivity: 0.95, Specificity: 0.27
 - Discitis Type
 - Spondilodiscitis Type: secondary, Sensitivity: 0.96, Specificity: 0.25
 - Estimated With Pseudocounts
 - MRI?
 - Spondilodiscitis Type: primary, Sensitivity: 0.95, Specificity: 0.12
 - Estimated With Pseudocounts
 - MRI?
 - Spondilodiscitis Type: secondary: surgery, Sensitivity: 0.90, Specificity: 0.38
 - Slightly more specific than in other groups
 - Spondilodiscitis Type: recidive, Sensitivity: 1.00, Specificity: 0.50
 - More specificity than in other cases!

Note: Redo analysis for PET before surgery only!

Distribution over bacteria

mibi_surgery_else	68	50.75%
mibi_surgery_CUTI	25	18.66%
mibi_surgery_STEP	22	16.42%
mibi_surgery_ECOL	11	8.21%
mibi_surgery_EFEC	5	3.73%
mibi_surgery_STAU	3	2.24%
	134	1

Non-spine foci analysis (which foci?)

- Compare with initialer Fokus if it makes sense
- Endocarditis: even if TEE is gold standard (PET does not replace TEE as it cannot reliable detect vegetations and therefore mostly native valve endocarditis!), PET detects sometimes
- Ortho infections detected → consequence on antibiotics in principle
 - Ortho (much more!), HNO, Dental, GI, Endocarditis, Uro more often in HWS
 - Pleuraempyem more often in BWS
 - Everything gets a little but rare to LWS
- Evaluate incidence of new foci per category of discitis!

Region	Focus	count	percent
--------	-------	-------	---------

HWS	Uro	2	5.71%
BWS	Uro	0	0.00%
LWS	Uro	0	0.00%
HWS	Soft tissue Infection	2	5.71%
BWS	Soft tissue Infection	3	5.26%
LWS	Soft tissue Infection	2	1.50%
LWS	Pneumonie	1	0.75%
HWS	Pneumonie	0	0.00%
BWS	Pneumonie	0	0.00%
BWS	Pleura	3	5.26%
HWS	Pleura	1	2.86%
LWS	Pleura	1	0.75%
HWS	Ortho	5	14.29%
BWS	Ortho	5	8.77%
LWS	Ortho	4	3.01%
HWS	HNO	2	5.71%
BWS	HNO	0	0.00%
LWS	HNO	0	0.00%
HWS	GI	1	2.86%
LWS	GI	2	1.50%
BWS	GI	0	0.00%
HWS	Endocarditis	1	2.86%
BWS	Endocarditis	1	1.75%
LWS	Endocarditis	2	1.50%
HWS	Dental	3	8.57%
LWS	Dental	9	6.77%
BWS	Dental	0	0.00%

Pending

PET and MRI

- We already argue that PET finds new foci, that is definitely an utility, especially in HWS Discitis.
- Here the question is: what is the utility of PET in cases when ignoring other nonspinal foci? Mostly in LWS: when is it possible to not use PET? In which subgroups (e.g. recidive, postoperative discitis)

Relationship with Psoas-abscesses and Epidural abscesses

- Check “intraspinal” column

- Exclude cases with epidural/psoas only and without spondylodiscitis!

Limit analysis to PET before surgery for spinal focus related analysis

Read the Gemini report and implement